Chapter 13

Vectors I

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13.1 Basic Formulas

A displacement of 3 km east is not the same object as the number 3. The number is a **scalar**: it has size only. The displacement is a **vector**: it has size and direction. Force, velocity and momentum are vectors. Distance, speed, mass and temperature are scalars.

> **What MF27 gives you.** The formula list has exactly two vector results: the **ratio theorem** (Section 13.1.4), and the **vector product in components** (Section 13.3). The scalar product, the equations of a line and of a plane, and every distance and angle formula in this chapter are *not* in MF27. They have to be remembered.

13.1.1 Notation and the Magnitude of a Vector

An arrow from P to Q is written $\overrightarrow{PQ}$. In print a vector is a bold letter, $\mathbf{a}$; in handwriting it is $\underline{a}$. The **zero vector 0** has length 0 and no direction. The vector $-\mathbf{a}$ has the same length as $\mathbf{a}$ and the opposite direction.

The unit vectors along the three axes are $\mathbf{i} = \begin{pmatrix}1\\0\\0\end{pmatrix}$, $\mathbf{j} = \begin{pmatrix}0\\1\\0\end{pmatrix}$ and $\mathbf{k} = \begin{pmatrix}0\\0\\1\end{pmatrix}$, so $\begin{pmatrix}x\\y\\z\end{pmatrix}$ and $x\mathbf{i}+y\mathbf{j}+z\mathbf{k}$ are the same vector. We use both.

Fix an origin O. The **position vector** of a point A is $\overrightarrow{OA}$: it says *where A is*. A **displacement vector** such as $\overrightarrow{AB}$ says how to get from A to B, and it does not care where it is drawn. If A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$, then

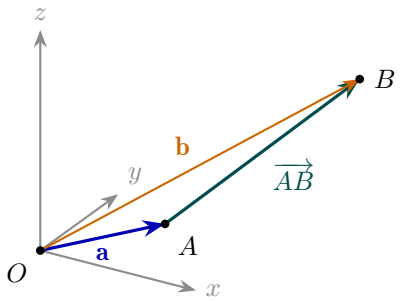
$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \mathbf{b} - \mathbf{a}.$$

Definition 13.1.1. Magnitude of a vector

Given a vector $\mathbf{a} = \begin{pmatrix}x\\y\\z\end{pmatrix}$, its **magnitude** (length) is

$$|\mathbf{a}| = \sqrt{x^2+y^2+z^2}.$$

The distance between A and B is $|\overrightarrow{AB}| = |\mathbf{b}-\mathbf{a}|$.



The picture uses $A(1,2,0)$ and $B(4,2,3)$ from the next example. The orange arrow is the position of B; the teal arrow is the displacement from A to B. They are different vectors, even though both end at B: $\overrightarrow{OA}+\overrightarrow{AB} = \overrightarrow{OB}$.

Example 13.1.1

Relative to an origin O, the points A and B have coordinates $(1, 2, 0)$ and $(4, 2, 3)$.

(a) Write $\overrightarrow{OA}$ and $\overrightarrow{AB}$ as column vectors.

(b) Find $|\overrightarrow{OB}|$ and the distance AB.

Solution. (a) $\overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$, $\overrightarrow{AB} = \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix}$.

(b) $|\overrightarrow{OB}| = \sqrt{4^2 + 2^2 + 3^2} = \sqrt{29}$, $AB = \sqrt{3^2 + 0^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$.

Common mistake: taking the distance AB as the difference of the two magnitudes. With $A(1, 2, 0)$ and $B(4, 2, 3)$:

$$AB = |\overrightarrow{OB}| - |\overrightarrow{OA}| = \sqrt{29} - \sqrt{5} \qquad \times$$

$$AB = |\overrightarrow{OB} - \overrightarrow{OA}| = \left| \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix} \right| = 3\sqrt{2} \qquad \checkmark$$

Subtract the vectors first, then take the magnitude.

Example 13.1.2

The point $P(2, k, -1)$ is at a distance of 3 units from $A(1, 2, 1)$. Find the possible values of k.

Solution. $\overrightarrow{AP} = \begin{pmatrix} 1 \\ k-2 \\ -2 \end{pmatrix}$, so $1 + (k-2)^2 + 4 = 9$, $(k-2)^2 = 4$, $k - 2 = \pm 2$, and $k = 0$ or 4.

$\overrightarrow{AB}$ and $\overrightarrow{BA}$ are not equal: $\overrightarrow{BA} = -\overrightarrow{AB}$. Always write "head minus tail": $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$, never the other way round.

13.1.2 Parallel Vectors and Proof of Collinearity

Definition 13.1.2. Parallel vectors

If $\mathbf{a}$ and $\mathbf{b}$ are non-zero parallel vectors, written $\mathbf{a} \parallel \mathbf{b}$, then

- $\mathbf{a} = k\mathbf{b}$, where k is a scalar constant, and
- $\mathbf{a} \times \mathbf{b} = \mathbf{0}$, where $\mathbf{0}$ is the zero vector (the vector product is in Section 13.3).

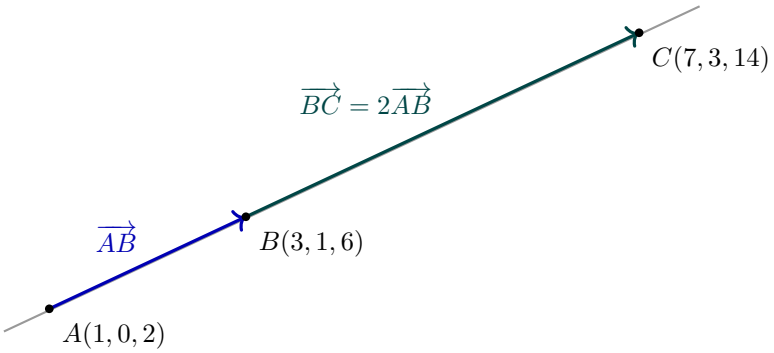
They point the same way if $k > 0$, and opposite ways if $k < 0$.

Definition 13.1.3. Proof of collinearity

To show that three points A, B and C are **collinear** (on one straight line), show any one of

$$\overrightarrow{AB} \parallel \overrightarrow{BC}, \quad \overrightarrow{AB} \parallel \overrightarrow{AC}, \quad \overrightarrow{BC} \parallel \overrightarrow{AC}, \quad \overrightarrow{BA} \parallel \overrightarrow{CB}, \quad \overrightarrow{BA} \parallel \overrightarrow{CA}, \quad \overrightarrow{CB} \parallel \overrightarrow{CA}.$$

Tip. Draw the diagram and take *any* two of the vectors that share a point. Two vectors that are parallel *and* share a point must lie on the same line.



Common mistake: showing that two vectors are parallel when they do *not* share a point, and concluding that A, B, C, D are collinear. With $A(0,0,0)$, $B(1,1,0)$, $C(0,1,0)$, $D(1,2,0)$:

$$\overrightarrow{AB} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \overrightarrow{CD}, \text{ so } A, B, C, D \text{ are collinear} \qquad \times$$

$\overrightarrow{AB} = \overrightarrow{CD}$ only says $AB \parallel CD$ (they are opposite sides of a parallelogram) ✓

For collinearity, the two vectors must share a point, as in each pair of the box above.

Example 13.1.3

Let $\mathbf{u} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

(a) Show that $\mathbf{v} = \begin{pmatrix} -6 \\ 3 \\ -6 \end{pmatrix}$ is parallel to $\mathbf{u}$, and find k such that $\mathbf{v} = k\mathbf{u}$. Explain why $\mathbf{w} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$ is not parallel to $\mathbf{u}$.

(b) Show that $A(1, 0, 2)$, $B(3, 1, 6)$ and $C(7, 3, 14)$ are collinear.

Solution. (a) $-3\mathbf{u} = \begin{pmatrix} -6 \\ 3 \\ -6 \end{pmatrix} = \mathbf{v}$, so $\mathbf{v} \parallel \mathbf{u}$ with $k = -3$. The two point opposite ways. If $\mathbf{w} = k\mathbf{u}$, the first component gives $k = 1$ and the third gives $k = \frac{1}{2}$. No single k works, so $\mathbf{w} \nparallel \mathbf{u}$.

(b) $\overrightarrow{AB} = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} 4 \\ 2 \\ 8 \end{pmatrix} = 2\overrightarrow{AB}$. So $\overrightarrow{AB} \parallel \overrightarrow{BC}$, and they share the point B. Hence A, B and C are collinear.

Example 13.1.4

The points $P(1, 2, 3)$, $Q(2, 4, 1)$ and $R(k, 8, m)$ are collinear. Find k and m.

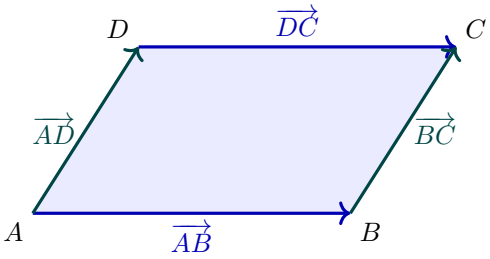
Solution. $\overrightarrow{PQ} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ and $\overrightarrow{PR} = \begin{pmatrix} k-1 \\ 6 \\ m-3 \end{pmatrix}$. Collinear, so $\overrightarrow{PR} = t\overrightarrow{PQ}$ for some t. The y-components give $6 = 2t$, so $t = 3$. Then $k - 1 = 3$ and $m - 3 = -6$, so $k = 4$ and $m = -3$.

13.1.3 Parallelograms and Unit Vectors

Definition 13.1.4. Properties of parallelograms

If $ABCD$ is a parallelogram (vertices in that order around it), then

$$\overrightarrow{AB} = \overrightarrow{DC}, \qquad \overrightarrow{AD} = \overrightarrow{BC}.$$



Both arrows of each pair must point the same way. $\overrightarrow{AB} = \overrightarrow{CD}$ is false: it is $-\overrightarrow{DC}$.

Definition 13.1.5. Unit vector

Let $\mathbf{a}$ be a non-zero vector. The **unit vector** of $\mathbf{a}$, written $\hat{\mathbf{a}}$, has magnitude 1 and the same direction as $\mathbf{a}$:

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}.$$

A vector of length L in the direction of $\mathbf{a}$ is $L\hat{\mathbf{a}}$. The unit vector is the tool that turns "direction" into "direction of length 1", which is exactly what the projection and distance formulas in Sections 13.2 and 13.3 need.

Example 13.1.5

$ABCD$ is a parallelogram with $A(1,2,0)$, $B(4,2,3)$ and $C(5,4,1)$. Find the coordinates of D.

Solution. $\overrightarrow{AD} = \overrightarrow{BC}$, so $\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{BC}$:

$$\overrightarrow{BC} = \begin{pmatrix}5\\4\\1\end{pmatrix} - \begin{pmatrix}4\\2\\3\end{pmatrix} = \begin{pmatrix}1\\2\\-2\end{pmatrix}, \qquad \overrightarrow{OD} = \begin{pmatrix}1\\2\\0\end{pmatrix} + \begin{pmatrix}1\\2\\-2\end{pmatrix} = \begin{pmatrix}2\\4\\-2\end{pmatrix}.$$

D is $\begin{pmatrix}2\\4\\-2\end{pmatrix}$, that is $D(2,4,-2)$. Check: $\overrightarrow{DC} = \begin{pmatrix}3\\0\\3\end{pmatrix} = \overrightarrow{AB}$.

Common mistake: using $\overrightarrow{AB} = \overrightarrow{CD}$, with the arrows the wrong way round.

$$\overrightarrow{OD} = \overrightarrow{OC} + \overrightarrow{AB} = \begin{pmatrix}5\\4\\1\end{pmatrix} + \begin{pmatrix}3\\0\\3\end{pmatrix} = \begin{pmatrix}8\\4\\4\end{pmatrix} \qquad \times$$

$$\overrightarrow{OD} = \overrightarrow{OC} - \overrightarrow{AB} = \begin{pmatrix}5\\4\\1\end{pmatrix} - \begin{pmatrix}3\\0\\3\end{pmatrix} = \begin{pmatrix}2\\4\\-2\end{pmatrix} \qquad \checkmark$$

$(8,4,4)$ makes $ABDC$ a parallelogram, with the vertices in the wrong order. Draw the diagram first.

Example 13.1.6

Let $\mathbf{a} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$. Find $\hat{\mathbf{a}}$, and the vector of magnitude 6 in the opposite direction to $\mathbf{a}$.

Solution. $|\mathbf{a}| = \sqrt{4+1+4} = 3$, so $\hat{\mathbf{a}} = \frac{1}{3}\begin{pmatrix}2\\1\\2\end{pmatrix} = \begin{pmatrix}\frac{2}{3}\\\frac{1}{3}\\\frac{2}{3}\end{pmatrix}$. The required vector is $-6\hat{\mathbf{a}} = \begin{pmatrix}-4\\-2\\-4\end{pmatrix}$. Check: $\sqrt{16+4+16} = 6$.

Common mistake: dividing by $|\mathbf{a}|^2$ (forgetting the square root) when making a unit vector.

$$\hat{\mathbf{a}} = \frac{1}{4+1+4}\begin{pmatrix}2\\1\\2\end{pmatrix} = \begin{pmatrix}\frac{2}{9}\\\frac{1}{9}\\\frac{2}{9}\end{pmatrix} \quad \times$$

$$\hat{\mathbf{a}} = \frac{1}{\sqrt{4+1+4}}\begin{pmatrix}2\\1\\2\end{pmatrix} = \begin{pmatrix}\frac{2}{3}\\\frac{1}{3}\\\frac{2}{3}\end{pmatrix} \quad \checkmark$$

Check that the answer has length 1: $\frac{4+1+4}{81} \neq 1$, but $\frac{4+1+4}{9} = 1$.

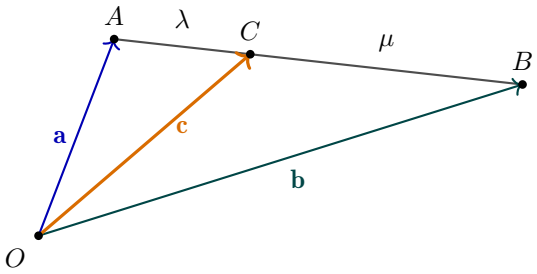
13.1.4 The Ratio Theorem

Definition 13.1.6. The ratio theorem (in MF27)

Let the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, and suppose that $\mu\, AC = \lambda\, CB$, that is, C divides AB in the ratio $AC : CB = \lambda : \mu$. Then

$$\mathbf{c} = \frac{\mu\mathbf{a} + \lambda\mathbf{b}}{\mu + \lambda}.$$

Each end is weighted by the part of the ratio *next to the other end*: the point closer to A gets more of $\mathbf{a}$. The midpoint ($\lambda = \mu = 1$) is $\dfrac{\mathbf{a}+\mathbf{b}}{2}$.



! **Read the ratio first, and draw the line.** The question may give $AB : CB$, or $3AC = 2CB$, or say "C lies on AB *produced*". Turn it into the form $AC : CB$ along the segment before you substitute. If C lies on AB produced, C is outside the segment: then B lies between A and C, and you apply the theorem with B as the dividing point and solve for $\mathbf{c}$.

Example 13.1.7

Let $\overrightarrow{OA} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\overrightarrow{OB} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$. Given that C lies on AB, find the coordinates of C if

(a) $AC : CB = 1 : 2$,

(b) $3AC = 2CB$,

(c) $2AB = 5CB$.

(d) Given instead that C lies on AB produced, and $3AC = 7BC$, find the coordinates of C.

Solution. (a) $\lambda : \mu = 1 : 2$, so $\overrightarrow{OC} = \dfrac{2\overrightarrow{OA} + \overrightarrow{OB}}{3} = \dfrac{1}{3}\begin{pmatrix}3\\4\\5\end{pmatrix}$, and C is $\begin{pmatrix}1\\\frac{4}{3}\\\frac{5}{3}\end{pmatrix}$, i.e. $C\left(1, \frac{4}{3}, \frac{5}{3}\right)$.

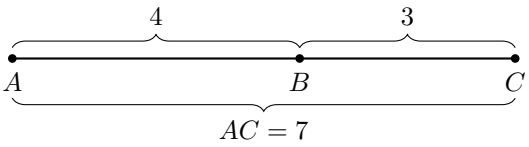
(b) $3AC = 2CB$ gives $AC : CB = 2 : 3$. So $\overrightarrow{OC} = \dfrac{3\overrightarrow{OA} + 2\overrightarrow{OB}}{5} = \dfrac{1}{5}\begin{pmatrix}5\\7\\9\end{pmatrix}$, and C is $\begin{pmatrix}1\\\frac{7}{5}\\\frac{9}{5}\end{pmatrix}$.

(c) $2AB = 5CB$ gives $AB : CB = 5 : 2$. Since C lies on AB, $AC = AB - CB$, so $AC : CB = 3 : 2$. Then $\overrightarrow{OC} = \dfrac{2\overrightarrow{OA} + 3\overrightarrow{OB}}{5} = \dfrac{1}{5}\begin{pmatrix}5\\8\\11\end{pmatrix}$, and C is $\begin{pmatrix}1\\\frac{8}{5}\\\frac{11}{5}\end{pmatrix}$.

(d) Draw the line: A, then B, then C. $AC : BC = 7 : 3$, so $AB : BC = 4 : 3$, and B divides AC in the ratio $4 : 3$:

$$\overrightarrow{OB} = \frac{3\overrightarrow{OA} + 4\overrightarrow{OC}}{7} \Rightarrow \overrightarrow{OC} = \frac{7\overrightarrow{OB} - 3\overrightarrow{OA}}{4} = \frac{1}{4}\begin{pmatrix} 4 \\ 11 \\ 18 \end{pmatrix},$$

and C is $\begin{pmatrix} 1 \\ \frac{11}{4} \\ \frac{9}{2} \end{pmatrix}$, i.e. $C\left(1, \frac{11}{4}, \frac{9}{2}\right)$.



Part (d): the diagram turns "$3AC = 7BC$, on AB produced" into "B divides AC in $4 : 3$".

Common mistake: reading $3AC = 2CB$ as $AC : CB = 3 : 2$. In part (b):

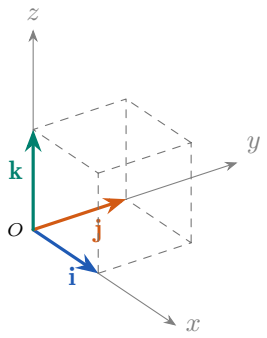
$$3AC = 2CB \Rightarrow AC : CB = 3 : 2, \quad \overrightarrow{OC} = \frac{2\overrightarrow{OA} + 3\overrightarrow{OB}}{5} = \begin{pmatrix} 1 \\ \frac{8}{5} \\ \frac{11}{5} \end{pmatrix} \quad \times$$

$$3AC = 2CB \Rightarrow \frac{AC}{CB} = \frac{2}{3}, \quad \overrightarrow{OC} = \frac{3\overrightarrow{OA} + 2\overrightarrow{OB}}{5} = \begin{pmatrix} 1 \\ \frac{7}{5} \\ \frac{9}{5} \end{pmatrix} \quad \checkmark$$

Divide both sides: $3AC = 2CB$ means AC is the *shorter* part. Check: C should then be nearer A.

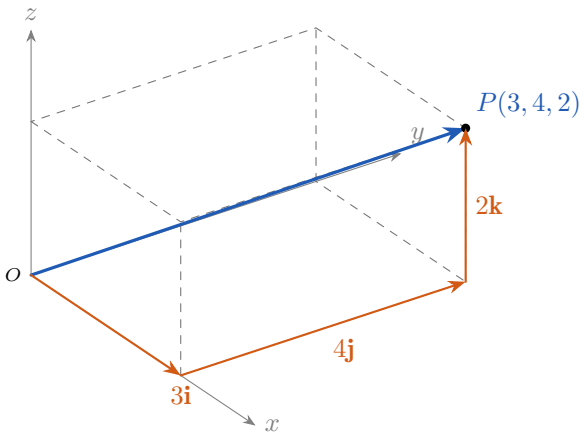
13.1.5 Visual Atlas: Basic Formulas

V1.1 Unit vectors i, j, k



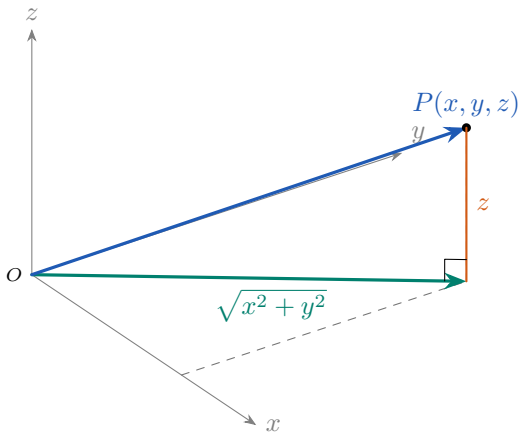
$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \qquad x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

V1.2 Position vector of $P(3, 4, 2)$



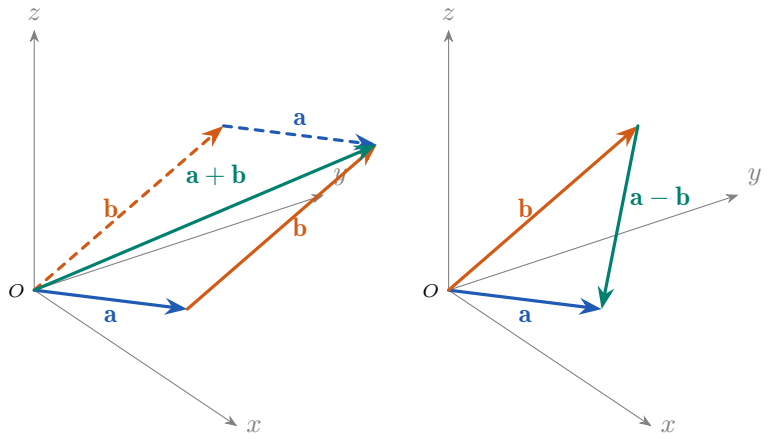
$$\overrightarrow{OP} = 3\mathbf{i} + 4\mathbf{j} + 2\mathbf{k} = \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix}$$

V1.3 Magnitude: Pythagoras twice



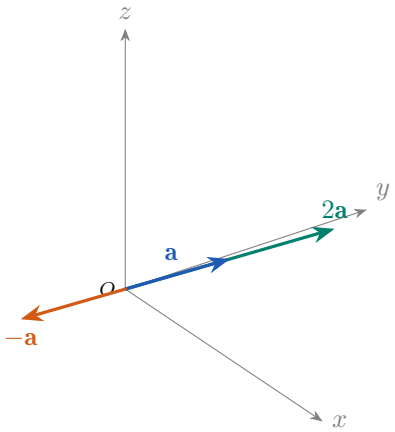
$$|\overrightarrow{OP}|^2 = \left(\sqrt{x^2+y^2}\right)^2 + z^2$$
$$|\overrightarrow{OP}| = \sqrt{x^2+y^2+z^2}$$

V1.4 Addition: triangle law and parallelogram law



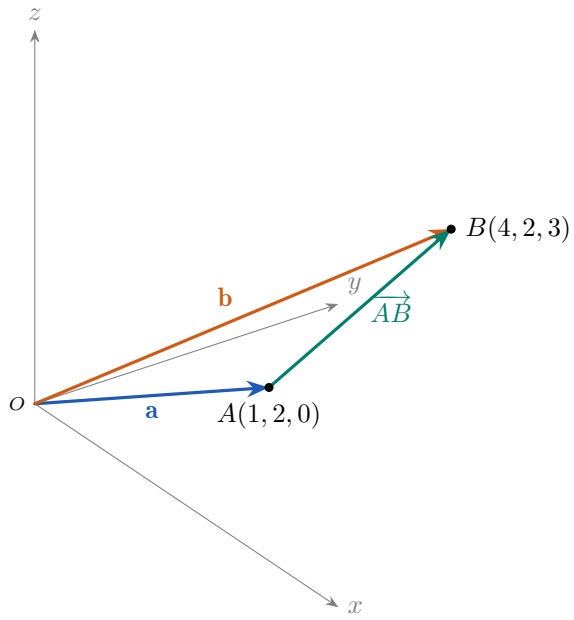
$$\mathbf{a}+\mathbf{b} = \mathbf{b}+\mathbf{a}$$
$$\overrightarrow{BA} = \mathbf{a}-\mathbf{b}$$

V1.5 Scalar multiples



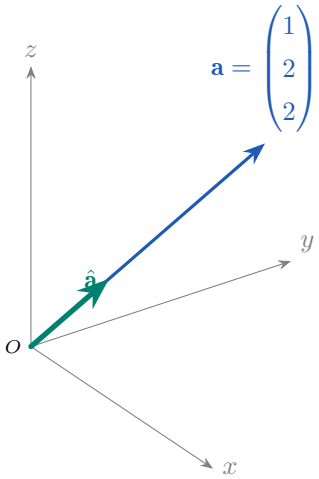
$$|k\mathbf{a}| = |k|\,|\mathbf{a}|$$

V1.6 Displacement $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$



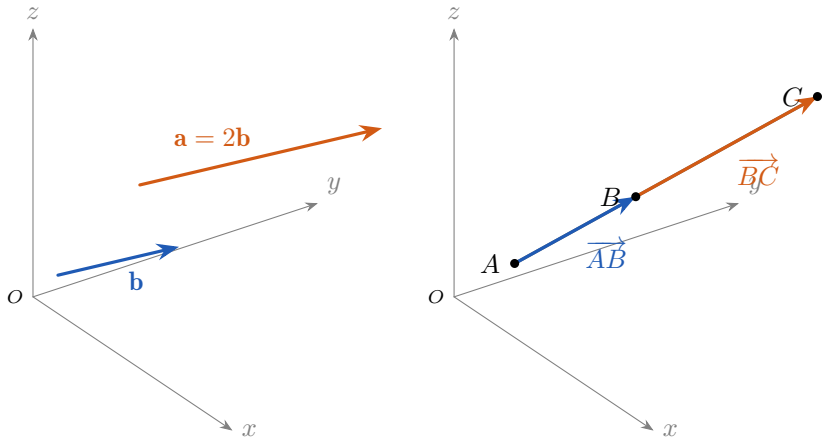
$$\begin{aligned}\overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\ &= \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix} \\ AB &= \sqrt{3^2 + 0^2 + 3^2} \\ &= 3\sqrt{2}\end{aligned}$$

V1.7 Unit vector



$$
\begin{aligned}
|\mathbf{a}| &= \sqrt{1+4+4} \\
&= 3 \\
\hat{\mathbf{a}} &= \frac{1}{3}\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}
\end{aligned}
$$

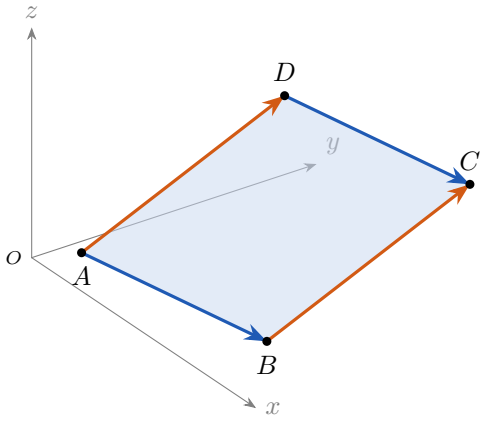
V1.8 Parallel vectors and collinear points



$$\mathbf{a} = k\mathbf{b} \iff \mathbf{a} \parallel \mathbf{b}$$

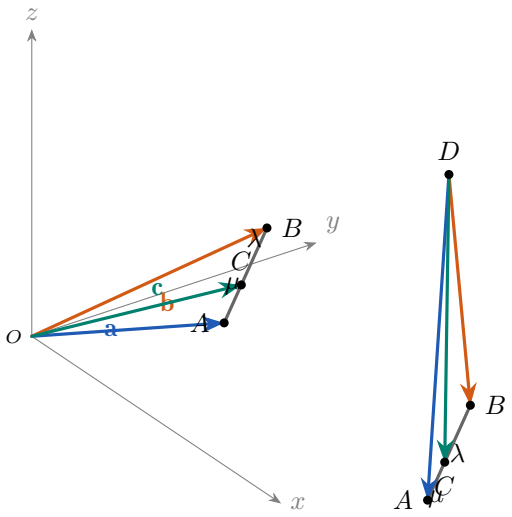
$\overrightarrow{AB} = k\overrightarrow{BC}$ and B common $\Rightarrow A, B, C$ collinear

V1.9 Parallelogram $ABCD$



$$
\begin{aligned}
\overrightarrow{AB} &= \overrightarrow{DC} \\
\overrightarrow{AD} &= \overrightarrow{BC} \\
\mathbf{d} &= \mathbf{a} + \mathbf{c} - \mathbf{b}
\end{aligned}
$$

V1.10 Ratio theorem



$$AC : CB = \mu : \lambda$$

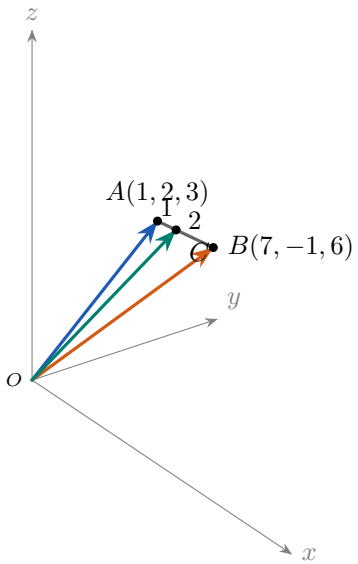
$$\mathbf{c} = \frac{\lambda\mathbf{a} + \mu\mathbf{b}}{\lambda + \mu}$$

$$\overrightarrow{DC} = \frac{\lambda\overrightarrow{DA} + \mu\overrightarrow{DB}}{\lambda + \mu}$$

$$\text{midpoint: } \mathbf{m} = \frac{\mathbf{a} + \mathbf{b}}{2}$$

Example 13.1.8

$A(1, 2, 3)$, $B(7, -1, 6)$. C on AB, $AC : CB = 1 : 2$. Find $\overrightarrow{OC}$.



Solution.

$$\begin{aligned}
\overrightarrow{OC} &= \frac{2\overrightarrow{OA} + \overrightarrow{OB}}{1+2} \\
&= \frac{1}{3}\left[2\begin{pmatrix}1\\2\\3\end{pmatrix} + \begin{pmatrix}7\\-1\\6\end{pmatrix}\right] \\
&= \frac{1}{3}\begin{pmatrix}9\\3\\12\end{pmatrix} \\
&= \begin{pmatrix}3\\1\\4\end{pmatrix}
\end{aligned}$$

13.2 The Scalar Product (Dot Product)

13.2.1 Definition and Properties of the Scalar Product

Definition 13.2.1. Scalar (dot) product

Given $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, the **scalar product** of $\mathbf{a}$ and $\mathbf{b}$ is

$$\mathbf{a} \cdot \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = a_1b_1 + a_2b_2 + a_3b_3.$$

The answer is a **scalar**, not a vector.

Definition 13.2.2. Properties of the scalar product

Let $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ be vectors and λ a scalar.

- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
- $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$
- $\lambda(\mathbf{a} \cdot \mathbf{b}) = (\lambda\mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (\lambda\mathbf{b})$
- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$

So a scalar product of two brackets expands like ordinary algebra, as long as every product is a dot between two vectors.

! **It is meaningless to write $\mathbf{a} \cdot \mathbf{b} \cdot \mathbf{c}$.** $\mathbf{a} \cdot \mathbf{b}$ is already a number, and a number cannot be dotted with the vector $\mathbf{c}$. Likewise $\mathbf{a}^2$ means nothing: write $\mathbf{a} \cdot \mathbf{a}$ or $|\mathbf{a}|^2$.

× **Common mistake:** the cancellation law does **not** hold for the dot product. Take $\mathbf{a} = \mathbf{i}$, $\mathbf{b} = \mathbf{j}$, $\mathbf{c} = \mathbf{k}$:

$$\mathbf{a} \cdot \mathbf{b} = 0 = \mathbf{a} \cdot \mathbf{c} \;\Rightarrow\; \mathbf{b} = \mathbf{c} \qquad \times$$

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c} \;\Rightarrow\; \mathbf{a} \cdot (\mathbf{b} - \mathbf{c}) = 0 \;\Rightarrow\; \mathbf{a} = \mathbf{0},\ \mathbf{b} = \mathbf{c},\ \text{or } \mathbf{a} \perp (\mathbf{b} - \mathbf{c}) \qquad \checkmark$$

Here $\mathbf{j} \neq \mathbf{k}$; it is the third case, $\mathbf{i} \perp (\mathbf{j} - \mathbf{k})$.

Example 13.2.1

Given that $|\mathbf{a}| = 3$, $|\mathbf{b}| = 2$ and $\mathbf{a} \cdot \mathbf{b} = 4$, find

(a) $(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} - 2\mathbf{b})$,

(b) $|\mathbf{a} + \mathbf{b}|$.

Solution. (a) Expand, keeping every product a dot product:

$$\begin{aligned}(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} - 2\mathbf{b}) &= \mathbf{a} \cdot \mathbf{a} - 2\,\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{a} - 2\,\mathbf{b} \cdot \mathbf{b} \\ &= |\mathbf{a}|^2 - \mathbf{a} \cdot \mathbf{b} - 2|\mathbf{b}|^2 = 9 - 4 - 8 = -3.\end{aligned}$$

(b) $|\mathbf{a} + \mathbf{b}|^2 = (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b}) = |\mathbf{a}|^2 + 2\,\mathbf{a} \cdot \mathbf{b} + |\mathbf{b}|^2 = 9 + 8 + 4 = 21$, so $|\mathbf{a} + \mathbf{b}| = \sqrt{21}$.

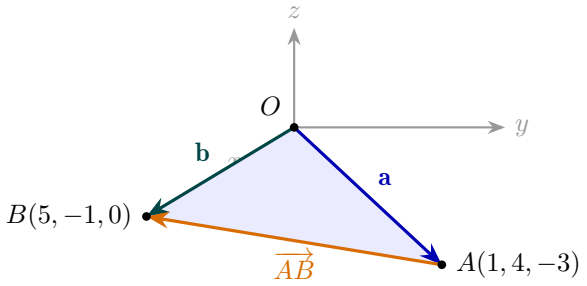
13.2.2 Angles: the Scalar Product Formula

Definition 13.2.3. The angle formula

Let $\mathbf{a}$ and $\mathbf{b}$ be the position vectors of A and B relative to the origin O, and let θ be the non-reflex angle AOB. Then

$$\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}|\,|\mathbf{b}|\cos\theta, \qquad \text{so} \qquad \cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}|\,|\mathbf{b}|}.$$

- If $\mathbf{a}\cdot\mathbf{b} > 0$, then θ is acute.
- If $\mathbf{a}\cdot\mathbf{b} < 0$, then θ is obtuse.
- If $\mathbf{a}\cdot\mathbf{b} = 0$, then θ is a right angle.



! **The two vectors must both point out of the angle, or both point in.** To find angle OAB, use $\overrightarrow{AO}$ and $\overrightarrow{AB}$ (both leave A). A "head-to-tail" pair such as $\overrightarrow{OA}$ and $\overrightarrow{AB}$ gives the supplement, $180° - \theta$, instead.

Example 13.2.2

Points O, A and B are such that $\overrightarrow{OA} = \mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$ and $\overrightarrow{OB} = 5\mathbf{i} - \mathbf{j}$.

(a) Find the size of angle AOB.

(b) Find the size of angle OAB.

Solution. (a) Both $\overrightarrow{OA}$ and $\overrightarrow{OB}$ leave O.

$$\cos\theta = \frac{\overrightarrow{OA}\cdot\overrightarrow{OB}}{|\overrightarrow{OA}|\,|\overrightarrow{OB}|} = \frac{(1)(5)+(4)(-1)+(-3)(0)}{\sqrt{1+16+9}\,\sqrt{25+1+0}} = \frac{1}{26}, \qquad \theta = 87.8° \text{ (1 d.p.)}.$$

(b) Use two vectors that leave A: $\overrightarrow{AO} = \begin{pmatrix}-1\\-4\\3\end{pmatrix}$ and $\overrightarrow{AB} = \begin{pmatrix}5\\-1\\0\end{pmatrix} - \begin{pmatrix}1\\4\\-3\end{pmatrix} = \begin{pmatrix}4\\-5\\3\end{pmatrix}$.

$$\cos\theta = \frac{-4+20+9}{\sqrt{26}\,\sqrt{50}} = \frac{25}{\sqrt{1300}}, \qquad \theta = 46.1° \text{ (1 d.p.)}.$$

× **Common mistake:** using the head-to-tail pair $\overrightarrow{OA}$ and $\overrightarrow{AB}$ for angle OAB.

$$\cos\theta = \frac{\overrightarrow{OA}\cdot\overrightarrow{AB}}{|\overrightarrow{OA}|\,|\overrightarrow{AB}|} = \frac{4-20-9}{\sqrt{26}\,\sqrt{50}} = \frac{-25}{\sqrt{1300}}, \quad \theta = 133.9° \qquad \times$$

$$\cos\theta = \frac{\overrightarrow{AO}\cdot\overrightarrow{AB}}{|\overrightarrow{AO}|\,|\overrightarrow{AB}|} = \frac{25}{\sqrt{1300}}, \quad \theta = 46.1° \qquad \checkmark$$

$133.9° + 46.1° = 180°$: the wrong pair measures the outside angle at A.

13.2.3 Perpendicularity of Vectors

Definition 13.2.4. Perpendicular vectors

Let $\mathbf{a}$ and $\mathbf{b}$ be non-zero vectors. Then

$$\mathbf{a} \cdot \mathbf{b} = 0 \iff \mathbf{a} \perp \mathbf{b}.$$

This is the case $\cos\theta = 0$ of the angle formula. It is the single most used fact in the chapter: every foot of a perpendicular, every normal to a plane, and every "show that … are perpendicular" comes back to a dot product equal to zero.

? If $\mathbf{a} \cdot \mathbf{b} = 0$, must one of $\mathbf{a}$ or $\mathbf{b}$ be the zero vector? What does your answer say about "cancelling" $\mathbf{a}$ in $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c}$?

Example 13.2.3

The vectors $\mathbf{a}$ and $\mathbf{b}$ are given by

$$\mathbf{a} = 2p\,\mathbf{i} + 3p\,\mathbf{j} + 6p\,\mathbf{k} \quad \text{and} \quad \mathbf{b} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k},$$

where $p > 0$. It is given that $|\mathbf{a}| = |\mathbf{b}|$.

(a) Find the exact value of p.

(b) Show that the vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$ are perpendicular.

Solution. (a) $|\mathbf{a}| = \sqrt{4p^2 + 9p^2 + 36p^2} = \sqrt{49p^2} = 7p$ (as $p > 0$), and $|\mathbf{b}| = \sqrt{1 + 4 + 4} = 3$. So $7p = 3$ and $p = \dfrac{3}{7}$.

(b) Expand with the properties of the dot product:

$$\begin{aligned}(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) &= \mathbf{a} \cdot \mathbf{a} - \mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{a} - \mathbf{b} \cdot \mathbf{b} \\ &= |\mathbf{a}|^2 - |\mathbf{b}|^2 = 0, \quad \text{since } |\mathbf{a}| = |\mathbf{b}|.\end{aligned}$$

Both vectors are non-zero, so $(\mathbf{a} + \mathbf{b}) \perp (\mathbf{a} - \mathbf{b})$.

i **Geometry behind (b).** $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$ are the two diagonals of the parallelogram on $\mathbf{a}$ and $\mathbf{b}$. When $|\mathbf{a}| = |\mathbf{b}|$ the parallelogram is a rhombus, and the diagonals of a rhombus are perpendicular. No components were needed, and none should be used: the question wants the algebra of the dot product.

Example 13.2.4

Find the value of k for which $\begin{pmatrix} 1 \\ k \\ 2 \end{pmatrix}$ is perpendicular to $\begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}$.

Solution. $\begin{pmatrix} 1 \\ k \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} = 3 + k - 4 = k - 1 = 0$, so $k = 1$.

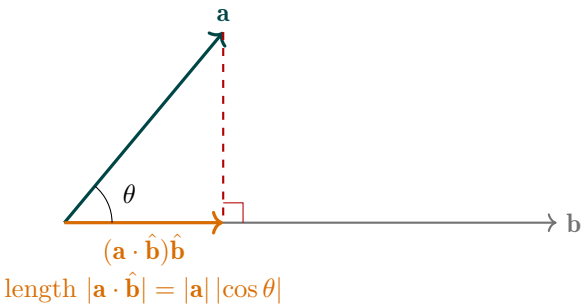
13.2.4 Projection

Definition 13.2.5. Projection of a onto b

Let $\mathbf{a}$ and $\mathbf{b}$ be vectors. Then

$$\text{Length of projection of } \mathbf{a} \text{ onto } \mathbf{b} = |\mathbf{a}\cdot\hat{\mathbf{b}}|, \qquad \text{Projection vector of } \mathbf{a} \text{ onto } \mathbf{b} = (\mathbf{a}\cdot\hat{\mathbf{b}})\,\hat{\mathbf{b}}.$$

The projection is the "shadow" of $\mathbf{a}$ on the line of $\mathbf{b}$. Only the *direction* of $\mathbf{b}$ matters, which is why it appears as the unit vector $\hat{\mathbf{b}}$. This is the geometrical meaning of $\mathbf{a}\cdot\hat{\mathbf{n}}$ that the syllabus asks for.



The order matters. The projection of $\mathbf{a}$ onto $\mathbf{b}$ is not the same as the projection of $\mathbf{b}$ onto $\mathbf{a}$: the lengths are $\dfrac{|\mathbf{a}\cdot\mathbf{b}|}{|\mathbf{b}|}$ and $\dfrac{|\mathbf{a}\cdot\mathbf{b}|}{|\mathbf{a}|}$. "Onto $\mathbf{b}$" means make $\mathbf{b}$ the unit vector.

Example 13.2.5

Given that $\mathbf{a} = 2\mathbf{i}+\mathbf{j}+2\mathbf{k}$ and $\mathbf{b} = -\mathbf{i}+2\mathbf{j}-\mathbf{k}$, find the length of projection and the projection vector of

(a) $\mathbf{a}$ onto $\mathbf{b}$,

(b) $\mathbf{b}$ onto $\mathbf{a}$.

Solution. $\mathbf{a}\cdot\mathbf{b} = -2+2-2 = -2$, $\quad|\mathbf{b}| = \sqrt{6}$, $\quad|\mathbf{a}| = 3$.

(a) $\mathbf{a}\cdot\hat{\mathbf{b}} = \dfrac{-2}{\sqrt{6}}$, so the length of projection is $\left|\dfrac{-2}{\sqrt{6}}\right| = \dfrac{2}{\sqrt{6}}$, and

$$\text{projection vector} = \left(\frac{-2}{\sqrt{6}}\right)\frac{1}{\sqrt{6}}\begin{pmatrix}-1\\2\\-1\end{pmatrix} = -\frac{1}{3}\begin{pmatrix}-1\\2\\-1\end{pmatrix} = \begin{pmatrix}\frac{1}{3}\\-\frac{2}{3}\\\frac{1}{3}\end{pmatrix}.$$

(b) $\mathbf{b}\cdot\hat{\mathbf{a}} = \dfrac{-2}{3}$, so the length of projection is $\dfrac{2}{3}$, and

$$\text{projection vector} = \left(-\frac{2}{3}\right)\frac{1}{3}\begin{pmatrix}2\\1\\2\end{pmatrix} = -\frac{2}{9}\begin{pmatrix}2\\1\\2\end{pmatrix} = \begin{pmatrix}-\frac{4}{9}\\-\frac{2}{9}\\-\frac{4}{9}\end{pmatrix}.$$

Common mistake: leaving the minus sign in a length, or forgetting to divide by $|\mathbf{b}|$.

$$\text{length of projection of } \mathbf{a} \text{ onto } \mathbf{b} = \mathbf{a}\cdot\mathbf{b} = -2 \qquad \times$$

$$\text{length of projection of } \mathbf{a} \text{ onto } \mathbf{b} = |\mathbf{a}\cdot\hat{\mathbf{b}}| = \frac{2}{\sqrt{6}} \qquad \checkmark$$

A length is never negative, and $\mathbf{b}$ must be made a unit vector first.

Check a projection vector. What is left over, $\mathbf{a}-(\mathbf{a}\cdot\hat{\mathbf{b}})\hat{\mathbf{b}}$, must be perpendicular to $\mathbf{b}$. In Example 13.2.5(a): $\begin{pmatrix}2\\1\\2\end{pmatrix} - \begin{pmatrix}\frac{1}{3}\\-\frac{2}{3}\\\frac{1}{3}\end{pmatrix} = \frac{5}{3}\begin{pmatrix}1\\1\\1\end{pmatrix}$, and $\begin{pmatrix}1\\1\\1\end{pmatrix}\cdot\begin{pmatrix}-1\\2\\-1\end{pmatrix} = 0$. ✓ This leftover is the vector $\mathbf{v}$ of Section 13.3.2.

Example 13.2.6

With reference to the origin O, the points A, B and C have position vectors

$$\mathbf{a} = \mathbf{i}+2\mathbf{j}-\mathbf{k}, \qquad \mathbf{b} = 3\mathbf{i}+\mathbf{k}, \qquad \mathbf{c} = 2\mathbf{i}+2\mathbf{j}+3\mathbf{k}.$$

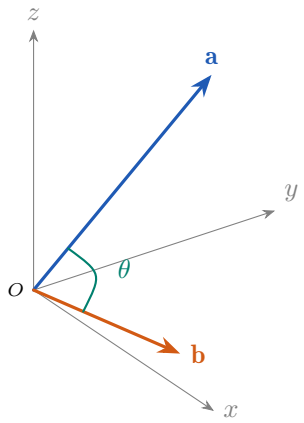
Find the length of projection of $\overrightarrow{BA}$ onto $\overrightarrow{BC}$.

Solution. $\overrightarrow{BA} = \begin{pmatrix}1\\2\\-1\end{pmatrix} - \begin{pmatrix}3\\0\\1\end{pmatrix} = \begin{pmatrix}-2\\2\\-2\end{pmatrix}$, $\overrightarrow{BC} = \begin{pmatrix}2\\2\\3\end{pmatrix} - \begin{pmatrix}3\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\2\\2\end{pmatrix}$, $|\overrightarrow{BC}| = 3$.

$$\text{Length of projection} = \left|\begin{pmatrix}-2\\2\\-2\end{pmatrix}\cdot\frac{1}{3}\begin{pmatrix}-1\\2\\2\end{pmatrix}\right| = \frac{|2+4-4|}{3} = \frac{2}{3} \text{ units.}$$

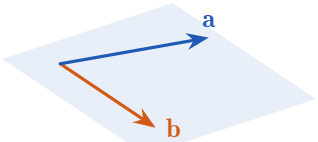
13.2.5 Visual Atlas: Scalar Product

V2.1 Angle θ: tails together

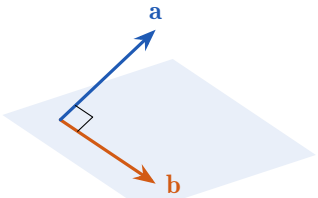


$$\mathbf{a}\cdot\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$$
$$\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$$
$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

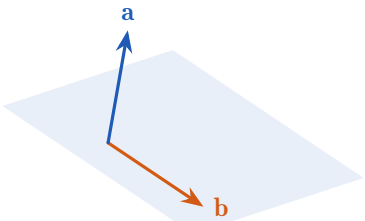
V2.2 Sign of $\mathbf{a}\cdot\mathbf{b}$: three cases



$\mathbf{a}\cdot\mathbf{b} > 0$, acute

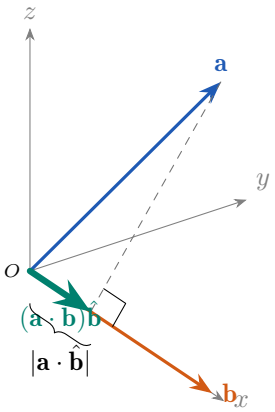


$\mathbf{a}\cdot\mathbf{b} = 0$, $\perp$



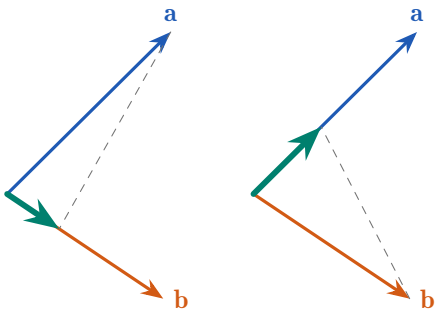
$\mathbf{a}\cdot\mathbf{b} < 0$, obtuse

V2.3 Length of projection and projection vector



$$\text{length of projection of } \mathbf{a} \text{ onto } \mathbf{b} = |\mathbf{a}\cdot\hat{\mathbf{b}}|$$
$$\text{projection vector} = (\mathbf{a}\cdot\hat{\mathbf{b}})\hat{\mathbf{b}}$$

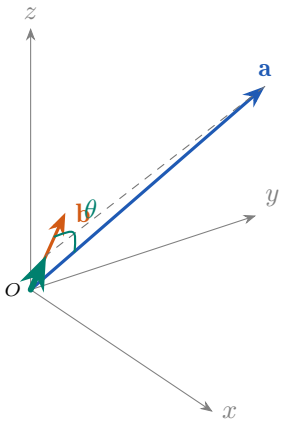
V2.4 Projection of a onto b $\neq$ projection of b onto a



$$|\mathbf{a}\cdot\hat{\mathbf{b}}| = \frac{|\mathbf{a}\cdot\mathbf{b}|}{|\mathbf{b}|}, \qquad |\mathbf{b}\cdot\hat{\mathbf{a}}| = \frac{|\mathbf{a}\cdot\mathbf{b}|}{|\mathbf{a}|}$$

Example 13.2.7

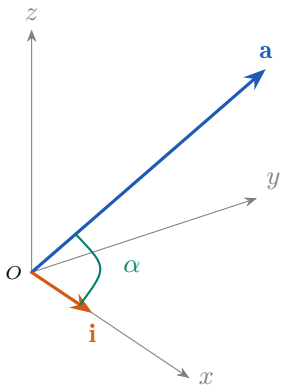
$\mathbf{a} = \begin{pmatrix}1\\2\\2\end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}2\\-1\\2\end{pmatrix}$. Find $\angle$ between $\mathbf{a}$ and $\mathbf{b}$, the length of projection of $\mathbf{a}$ onto $\mathbf{b}$, and the projection vector.



Solution.

$$\begin{aligned}
\mathbf{a}\cdot\mathbf{b} &= 2-2+4\\
&= 4\\
|\mathbf{a}| = 3, \quad |\mathbf{b}| &= 3\\
\cos\theta &= \frac{4}{3(3)}\\
\theta &= \cos^{-1}\frac{4}{9}\\
&= 63.6^\circ \text{ (1 d.p.)}\\
|\mathbf{a}\cdot\hat{\mathbf{b}}| &= \frac{4}{3}\\
(\mathbf{a}\cdot\hat{\mathbf{b}})\hat{\mathbf{b}} &= \frac{4}{3}\cdot\frac{1}{3}\begin{pmatrix}2\\-1\\2\end{pmatrix}\\
&= \frac{4}{9}\begin{pmatrix}2\\-1\\2\end{pmatrix}
\end{aligned}$$

V2.5 Angle with an axis



$$\begin{aligned}
\cos\alpha &= \frac{\mathbf{a}\cdot\mathbf{i}}{|\mathbf{a}||\mathbf{i}|}\\
&= \frac{1}{3}\\
\alpha &= 70.5^\circ \text{ (1 d.p.)}
\end{aligned}$$

13.3 The Vector Product (Cross Product)

13.3.1 Definition and Properties of the Vector Product

Definition 13.3.1. Vector (cross) product, in MF27

Given $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, the **vector product** of $\mathbf{a}$ and $\mathbf{b}$ is

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ -(a_1b_3 - a_3b_1) \\ a_1b_2 - a_2b_1 \end{pmatrix}.$$

The answer is a **vector**.

How to remember it. For each row, cover that row and take the "cross" of the two rows left, top-left times bottom-right minus top-right times bottom-left. Then put a minus sign in front of the *middle* row only. The middle component is the one most often written with the wrong sign.

Definition 13.3.2. Properties of the vector product

Let $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ be vectors and λ a scalar.

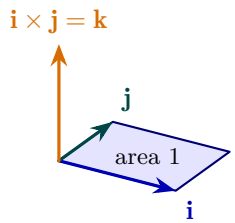
- $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$ (it is *not* commutative)
- $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$
- $\lambda(\mathbf{a} \times \mathbf{b}) = (\lambda\mathbf{a}) \times \mathbf{b} = \mathbf{a} \times (\lambda\mathbf{b})$
- $\mathbf{a} \times \mathbf{a} = \mathbf{0}$

As with the dot product, the **cancellation law does not hold**: $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$ does not give $\mathbf{b} = \mathbf{c}$.

Definition 13.3.3. What the vector product tells you

Let $\mathbf{a}$ and $\mathbf{b}$ be non-zero vectors, and θ the non-reflex angle between them.

- $\mathbf{a} \times \mathbf{b}$ is **perpendicular to both** $\mathbf{a}$ and $\mathbf{b}$.
- $\mathbf{a} \times \mathbf{b} = \mathbf{0} \iff \mathbf{a} \parallel \mathbf{b}$. (Section 13.1.2 gives the other test, $\mathbf{a} = k\mathbf{b}$.)
- $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}| \sin\theta$.



The direction is given by the right-hand rule: fingers from $\mathbf{a}$ towards $\mathbf{b}$, thumb along $\mathbf{a} \times \mathbf{b}$. Swapping the order flips the direction but not the length, so an area does not care about the order, and $\mathbf{n}$ and $-\mathbf{n}$ are normals to the same plane.

?

$|\mathbf{a}\times\mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}|\sin\theta$ can also be used to find the angle between two vectors, but it is usually undesirable. If $\sin\theta = \frac{1}{2}$, what are the two possible values of θ? Which formula, the dot or the cross, can tell them apart?

Example 13.3.1

(a) Find $\mathbf{a}\times\mathbf{b}$ for $\mathbf{a} = \begin{pmatrix}1\\2\\2\end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix}2\\-2\\1\end{pmatrix}$, and check it is perpendicular to both.

(b) Use the vector product to show that $\begin{pmatrix}2\\-1\\3\end{pmatrix}$ and $\begin{pmatrix}-4\\2\\-6\end{pmatrix}$ are parallel.

Solution. (a) $\mathbf{a}\times\mathbf{b} = \begin{pmatrix}2\cdot 1 - 2\cdot(-2)\\-(1\cdot 1 - 2\cdot 2)\\1\cdot(-2) - 2\cdot 2\end{pmatrix} = \begin{pmatrix}6\\3\\-6\end{pmatrix}$. Check: $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{a} = 6+6-12 = 0$ and $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{b} = 12-6-6 = 0$.

(b) $\begin{pmatrix}2\\-1\\3\end{pmatrix}\times\begin{pmatrix}-4\\2\\-6\end{pmatrix} = \begin{pmatrix}6-6\\-(-12+12)\\4-4\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$, so the vectors are parallel.

×

Common mistake: forgetting the minus sign on the middle component. In (a):

$$\mathbf{a}\times\mathbf{b} = \begin{pmatrix}2\cdot 1 - 2\cdot(-2)\\1\cdot 1 - 2\cdot 2\\1\cdot(-2) - 2\cdot 2\end{pmatrix} = \begin{pmatrix}6\\-3\\-6\end{pmatrix} \quad \times$$

$$\mathbf{a}\times\mathbf{b} = \begin{pmatrix}2\cdot 1 - 2\cdot(-2)\\-(1\cdot 1 - 2\cdot 2)\\1\cdot(-2) - 2\cdot 2\end{pmatrix} = \begin{pmatrix}6\\3\\-6\end{pmatrix} \quad \checkmark$$

The perpendicularity check catches it: $\begin{pmatrix}6\\-3\\-6\end{pmatrix}\cdot\begin{pmatrix}1\\2\\2\end{pmatrix} = -12 \neq 0$.

13.3.2 Perpendicular Distance

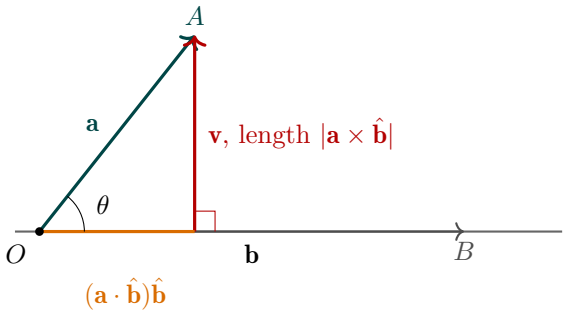
Definition 13.3.4. Perpendicular distance from a point to a line through O

Let $\mathbf{a}$ and $\mathbf{b}$ be the position vectors of A and B. The perpendicular distance from A to the line through O and B is

$$\text{Perpendicular distance} = |\mathbf{a} \times \hat{\mathbf{b}}|,$$

and the associated vector (from the foot of the perpendicular to A) is

$$\mathbf{v} = \mathbf{a} - (\mathbf{a} \cdot \hat{\mathbf{b}})\,\hat{\mathbf{b}}.$$



In the right-angled triangle, $|\mathbf{a}|$ is the hypotenuse and θ is the angle at O. The side *along* $\mathbf{b}$ is $|\mathbf{a}|\cos\theta = |\mathbf{a} \cdot \hat{\mathbf{b}}|$ (the projection), and the side *perpendicular* to $\mathbf{b}$ is $|\mathbf{a}|\sin\theta = |\mathbf{a} \times \hat{\mathbf{b}}|$. The vector $\mathbf{v}$ is $\mathbf{a}$ minus its projection.

! This is not the same as the perpendicular distance from B to the line joining O and A. The point goes in the first slot, and the **line's direction** is the one made a unit vector. If the line does not pass through O, replace $\mathbf{a}$ by a vector from any point on the line to the point (Section 13.4.4).

Example 13.3.2

With reference to the origin O, the points A, B and C have position vectors

$$\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}, \qquad \mathbf{b} = \mathbf{i} - \mathbf{j} + \mathbf{k}, \qquad \mathbf{c} = 3\mathbf{i} - \mathbf{j} + \mathbf{k}.$$

Find the shortest distance from C to the line joining A and B.

Solution. Take A as the point on the line. Then $\overrightarrow{AC}$ plays the role of $\mathbf{a}$ and $\overrightarrow{AB}$ the role of $\mathbf{b}$:

$$\overrightarrow{AB} = \begin{pmatrix} 0 \\ -3 \\ -2 \end{pmatrix}, \qquad \overrightarrow{AC} = \begin{pmatrix} 2 \\ -3 \\ -2 \end{pmatrix}, \qquad |\overrightarrow{AB}| = \sqrt{13}.$$

$$\overrightarrow{AC} \times \overrightarrow{AB} = \begin{pmatrix} (-3)(-2) - (-2)(-3) \\ -\big(2(-2) - (-2)(0)\big) \\ 2(-3) - (-3)(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ -6 \end{pmatrix}.$$

$$\text{Distance} = \left|\overrightarrow{AC} \times \widehat{\overrightarrow{AB}}\right| = \frac{\sqrt{0 + 16 + 36}}{\sqrt{13}} = \frac{\sqrt{52}}{\sqrt{13}} = 2 \text{ units}.$$

Check: $\overrightarrow{BC} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$ and $\overrightarrow{BC} \cdot \overrightarrow{AB} = 0$, so the foot of the perpendicular is B itself, and $BC = 2$.

Common mistake: using $\overrightarrow{AB}$ instead of the *unit* vector, so the answer is stretched by $|\overrightarrow{AB}|$.

$$\text{Distance} = \left|\overrightarrow{AC} \times \overrightarrow{AB}\right| = \sqrt{52} = 2\sqrt{13} \qquad \times$$

$$\text{Distance} = \left|\overrightarrow{AC} \times \widehat{\overrightarrow{AB}}\right| = \frac{\sqrt{52}}{\sqrt{13}} = 2 \qquad \checkmark$$

$2\sqrt{13}$ is the *area of the parallelogram* on $\overrightarrow{AB}$ and $\overrightarrow{AC}$ (Section 13.3.3), not a length.

Example 13.3.3

Let $\mathbf{p} = \begin{pmatrix} 3 \\ 0 \\ 6 \end{pmatrix}$ and $\hat{\mathbf{n}} = \begin{pmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}$, a unit vector. Find $\mathbf{p} \cdot \hat{\mathbf{n}}$ and $|\mathbf{p} \times \hat{\mathbf{n}}|$, say what each measures, and check Pythagoras with $|\mathbf{p}|$.

Solution. $\mathbf{p} \cdot \hat{\mathbf{n}} = \frac{1}{3}(6 + 0 + 6) = 4$: the length of the projection of $\mathbf{p}$ along $\hat{\mathbf{n}}$.

$\mathbf{p} \times \hat{\mathbf{n}} = \frac{1}{3}\begin{pmatrix} 0 - 12 \\ -(3 - 12) \\ 6 - 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \\ 2 \end{pmatrix}$, so $|\mathbf{p} \times \hat{\mathbf{n}}| = \sqrt{29}$: the perpendicular distance from P to the line through O in the direction $\hat{\mathbf{n}}$.

Check: $4^2 + 29 = 45 = 3^2 + 0^2 + 6^2 = |\mathbf{p}|^2$.

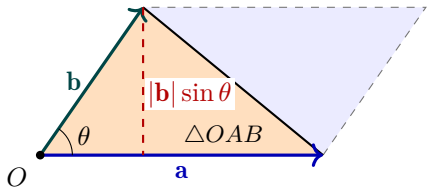
13.3.3 Areas of Triangles and Parallelograms

Definition 13.3.5. Areas

Let $\mathbf{a}$ and $\mathbf{b}$ be the position vectors of A and B relative to O. Then

$$\text{Area of } \triangle OAB = \frac{1}{2}|\mathbf{a}\times\mathbf{b}|, \qquad \text{Area of parallelogram with sides } \mathbf{a} \text{ and } \mathbf{b} = |\mathbf{a}\times\mathbf{b}|.$$

For a triangle without the origin as a vertex, use two sides from one vertex: Area of $\triangle ABC = \frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|$.



Base $|\mathbf{a}|$ times height $|\mathbf{b}|\sin\theta$ is $|\mathbf{a}|\,|\mathbf{b}|\sin\theta = |\mathbf{a}\times\mathbf{b}|$: the whole parallelogram. The triangle is half of it.

Example 13.3.4

(a) Find the area of the parallelogram with adjacent sides $\mathbf{a} = \begin{pmatrix}1\\2\\2\end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix}2\\-2\\1\end{pmatrix}$, and the area of triangle OAB.

(b) Using the points $A(1,2,3)$, $B(1,-1,1)$ and $C(3,-1,1)$ of Example 13.3.2, find the area of triangle ABC, and check it against $\frac{1}{2}\times$ base $\times$ height.

Solution. (a) From Example 13.3.1, $\mathbf{a}\times\mathbf{b} = 6\mathbf{i}+3\mathbf{j}-6\mathbf{k}$, so the parallelogram has area $\sqrt{36+9+36} = 9$ and the triangle has area $\frac{9}{2}$.

(b) $\overrightarrow{AB} = \begin{pmatrix}0\\-3\\-2\end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix}2\\-3\\-2\end{pmatrix}$, so $\overrightarrow{AB}\times\overrightarrow{AC} = \begin{pmatrix}6-6\\-(0+4)\\0+6\end{pmatrix} = \begin{pmatrix}0\\-4\\6\end{pmatrix}$ and

$$\text{Area of } \triangle ABC = \frac{1}{2}\sqrt{0+16+36} = \frac{1}{2}\sqrt{52} = \sqrt{13}.$$

Check: with base $AB = \sqrt{13}$, the height is the distance from C to line AB, which is 2 (Example 13.3.2). $\frac{1}{2}\times\sqrt{13}\times 2 = \sqrt{13}$.

Common mistake: using position vectors for a triangle that does not have O as a vertex.

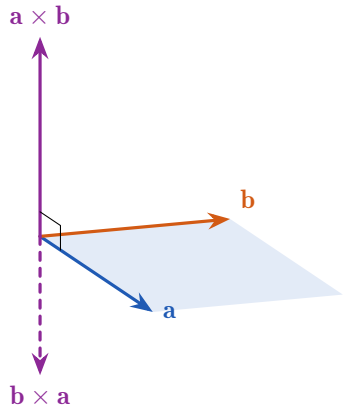
$$\text{Area of } \triangle ABC = \frac{1}{2}\left|\overrightarrow{OA}\times\overrightarrow{OB}\right| = \frac{1}{2}\left|\begin{pmatrix}5\\2\\-3\end{pmatrix}\right| = \frac{\sqrt{38}}{2} \qquad \times$$

$$\text{Area of } \triangle ABC = \frac{1}{2}\left|\overrightarrow{AB}\times\overrightarrow{AC}\right| = \sqrt{13} \qquad \checkmark$$

$\frac{1}{2}|\overrightarrow{OA}\times\overrightarrow{OB}|$ is the area of triangle OAB, a different triangle.

13.3.4 Visual Atlas: Vector Product

V3.1 $\mathbf{a} \times \mathbf{b}$ is perpendicular to both



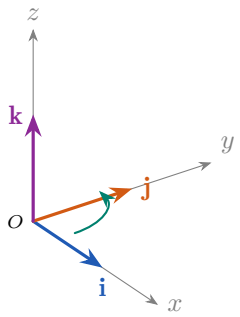
$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ -(a_1b_3 - a_3b_1) \\ a_1b_2 - a_2b_1 \end{pmatrix}$$

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = 0$$

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{b} = 0$$

$$\mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$$

V3.2 Right-hand rule: $\mathbf{i} \times \mathbf{j} = \mathbf{k}$



$$\mathbf{i} \times \mathbf{j} = \mathbf{k}, \qquad \mathbf{j} \times \mathbf{k} = \mathbf{i}, \qquad \mathbf{k} \times \mathbf{i} = \mathbf{j}$$

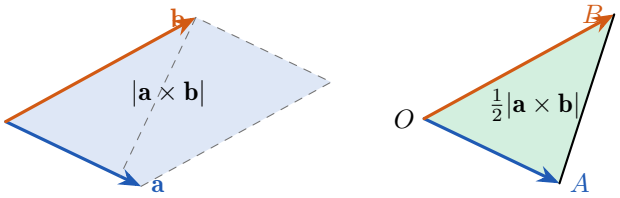
V3.3 Parallel vectors: $\mathbf{a} \times \mathbf{b} = \mathbf{0}$



$$\text{area} = 0$$

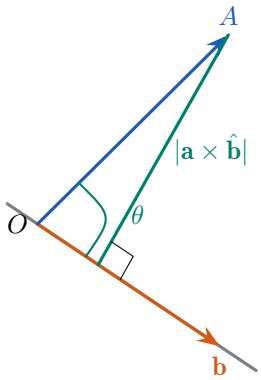
$$\theta = 0 \text{ or } 180^\circ \;\Rightarrow\; |\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta = 0$$

V3.4 Area of parallelogram and triangle



$$\begin{aligned}\text{Area of parallelogram} &= |\mathbf{a}||\mathbf{b}|\sin\theta \\ &= |\mathbf{a}\times\mathbf{b}| \\ \text{Area of } \triangle OAB &= \frac{1}{2}|\mathbf{a}\times\mathbf{b}| \\ \text{Area of } \triangle ABC &= \frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|\end{aligned}$$

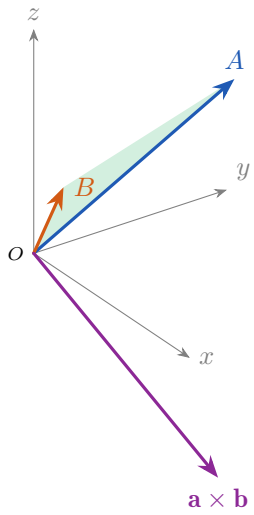
V3.5 Perpendicular distance from A to line OB



$$\begin{aligned}AF &= |\mathbf{a}|\sin\theta \\ &= \frac{|\mathbf{a}\times\mathbf{b}|}{|\mathbf{b}|} \\ &= |\mathbf{a}\times\hat{\mathbf{b}}|\end{aligned}$$

Example 13.3.5

$\mathbf{a} = \begin{pmatrix}1\\2\\2\end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}2\\-1\\2\end{pmatrix}$. Find $\mathbf{a}\times\mathbf{b}$, the area of $\triangle OAB$ and the perpendicular distance from A to OB.



Solution.

$$\begin{aligned}
\mathbf{a} \times \mathbf{b} &= \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \\
&= \begin{pmatrix} 2(2) - 2(-1) \\ -(1(2) - 2(2)) \\ 1(-1) - 2(2) \end{pmatrix} \\
&= \begin{pmatrix} 6 \\ 2 \\ -5 \end{pmatrix} \\
|\mathbf{a} \times \mathbf{b}| &= \sqrt{36 + 4 + 25} \\
&= \sqrt{65} \\
\text{Area of } \triangle OAB &= \frac{\sqrt{65}}{2} \\
\text{Distance} &= \frac{|\mathbf{a} \times \mathbf{b}|}{|\mathbf{b}|} \\
&= \frac{\sqrt{65}}{3}
\end{aligned}$$